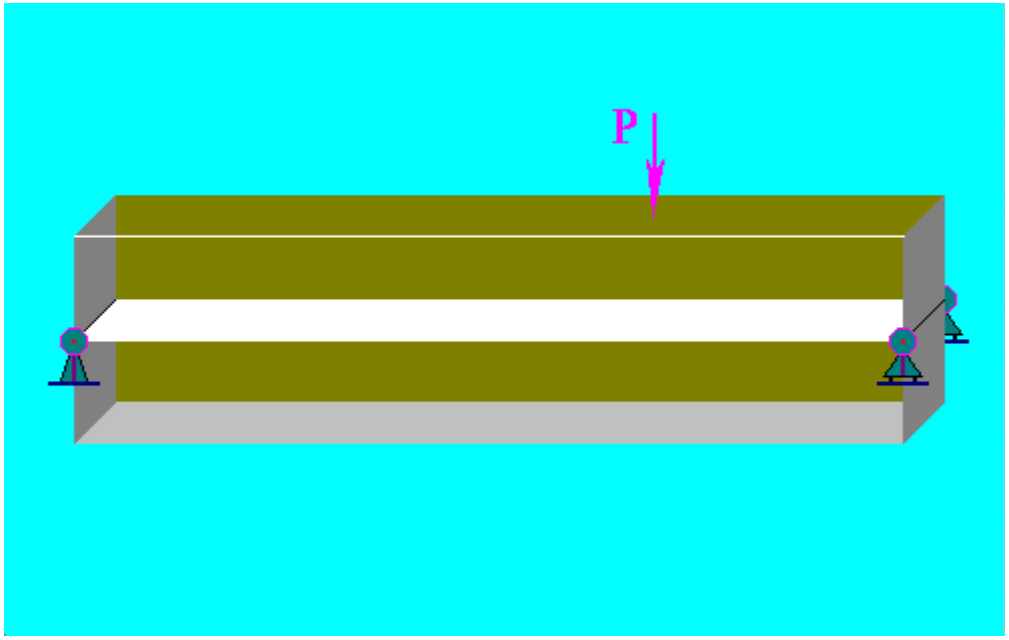
Chapter 6 Bending moment and shear force diagrams in beam

6.1 Introduction

We have looked at the stress and deformation of components under an axial load and when being twist. However, experience in the real world indicates that stresses are much higher when a component is being bent.

The first step in determining stresses in a beam being bent is to determine the internal bending moment and shear force being carried.

The procedures for constructing bending moment (BM) and shear force (SF) diagram were presented in the unit *Fundamentals of Mechanics*. In this chapter we shall use some examples to review these.



6.2 Sign convention for *internal forces* (shear force and bending moment)

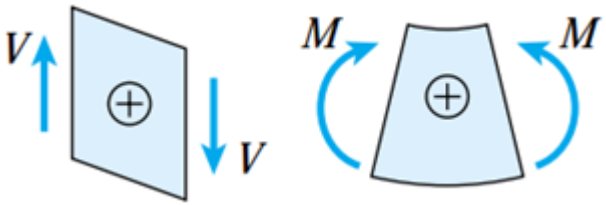


Shear force in beam

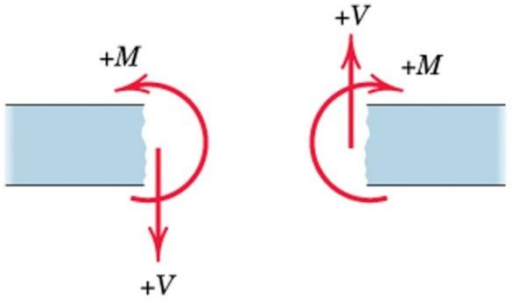


Bending moment in beam

It is customary to assume that the positive shear force tends to rotate the material clockwise and the bending moment tends to compress the upper part of the beam and elongate the lower part.



Positive forces and moments and the resulting deformations are shown acting on an element of a beam cut out between two cross sections that are a small distance apart.



Positive internal shear force and positive internal moment

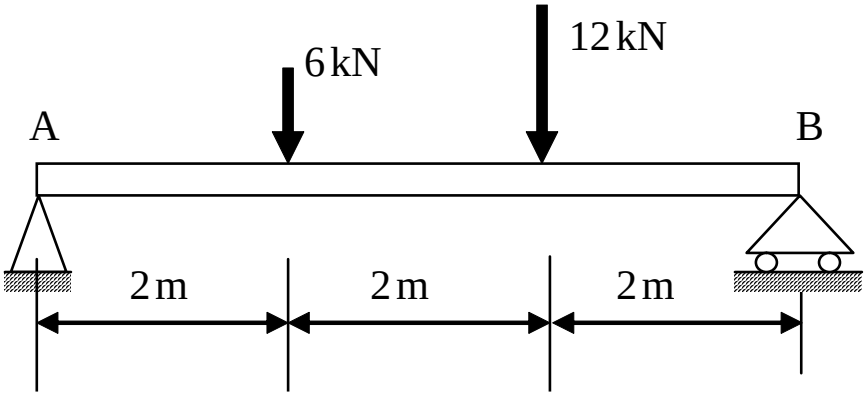
6.3 BM & SF Diagrams by using free body diagram

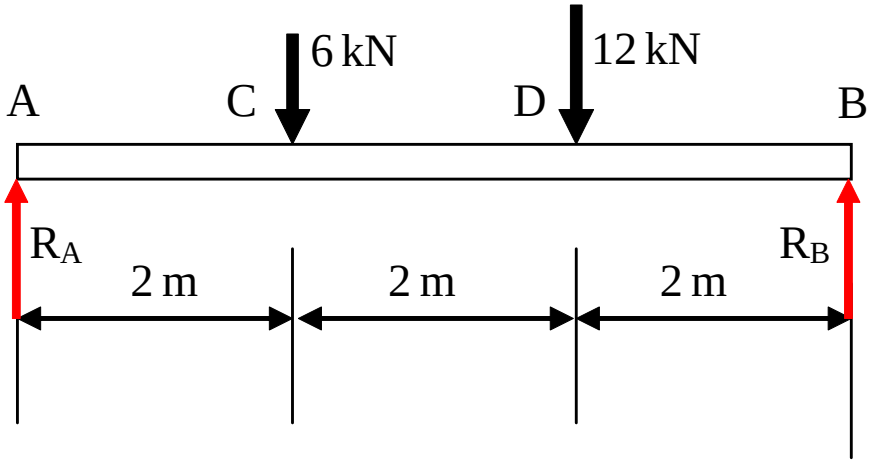


The internal forces can be obtained based on the following equilibrium conditions of the forces (including moments)

$$\uparrow \sum F = 0 , \quad \circlearrowleft \sum M = 0$$

Example 1: Draw the SF diagram and BM diagram for the beam shown. Find the location of the maximum bending moment.





Solution:

1 Find the support reactions

$\circlearrowleft \sum M_{\text{about point A}} = 0$

$\Rightarrow R_B \times 6 - 6 \times 2 - 12 \times 4 = 0$

$\Rightarrow R_B = 10 \text{ kN}$

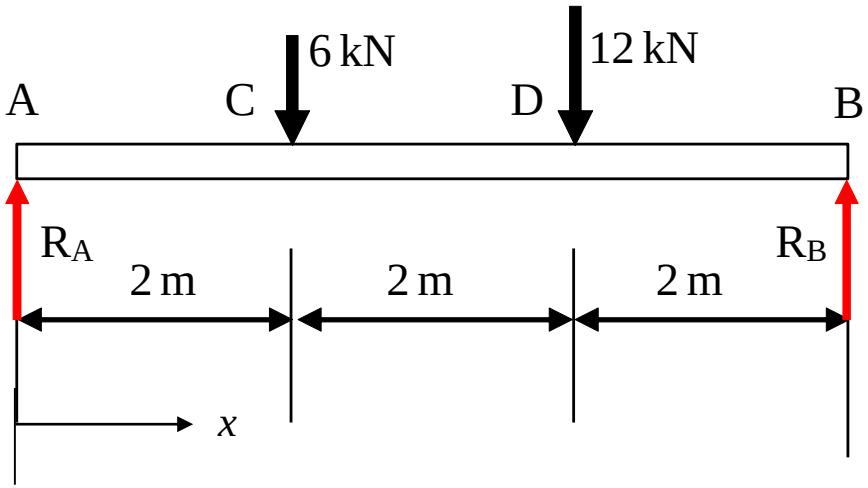
$\uparrow \sum F = 0$

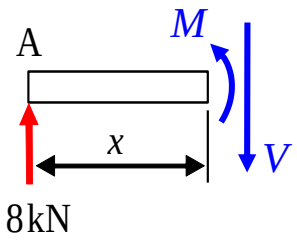
$\Rightarrow R_A + R_B - 12 - 6 = 0$

$\Rightarrow R_A = 8 \text{ kN}$

2 Divide the beam into 3 regions, based on the loading conditions: AC, CD & DB.

Expressing the SF and BM for each region.





Free body diagram

Note: when you study a free body, *V* and *M* act as external forces for this free body.

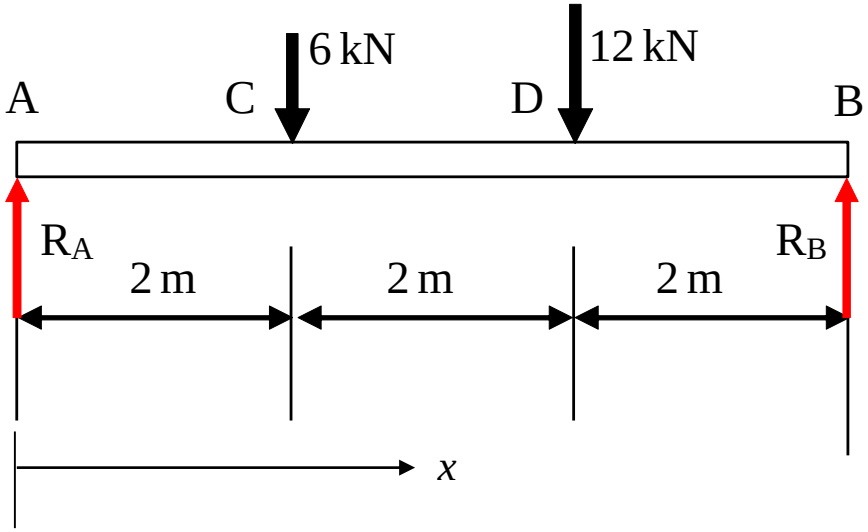
Region AC ($0 < x < 2$):

$\uparrow \sum F = 0$

$\Rightarrow 8 - V = 0 \Rightarrow V = 8\,\text{kN}$

$\circlearrowright \sum M_{\text{about location } x} = 0$

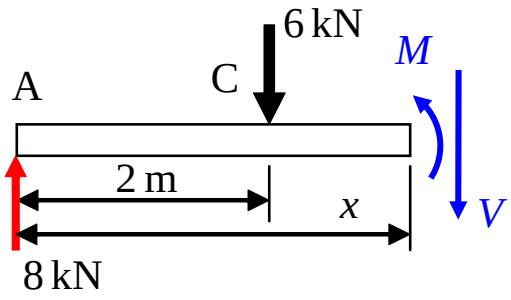
$\Rightarrow M - 8 \times x = 0 \Rightarrow M = 8x$



Region CD ($2 < x < 4$):

$\uparrow \sum F = 0$

$\Rightarrow 8 - 6 - V = 0 \Rightarrow V = 2\,\text{kN}$

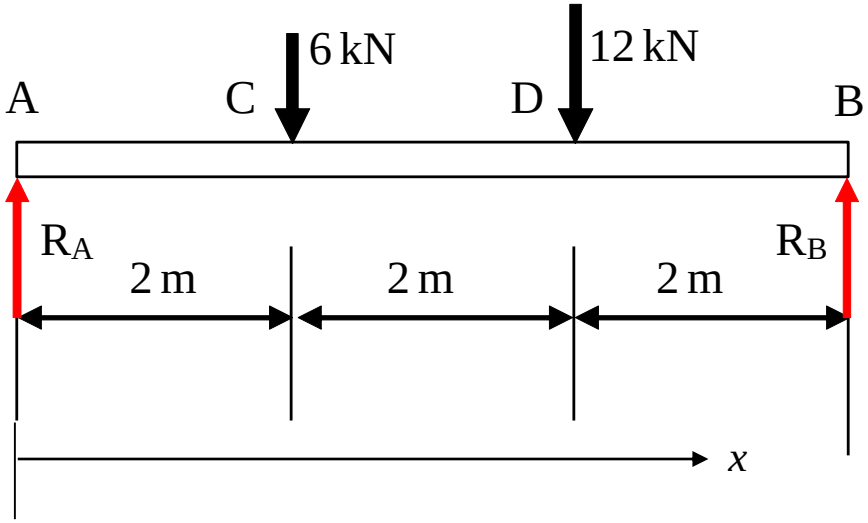


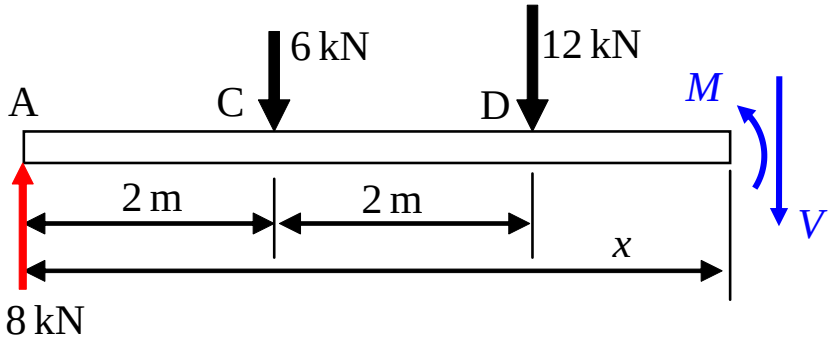
Free body diagram

$\circlearrowleft \sum M_{\text{about location } x} = 0$

$\Rightarrow M + 6\times(x-2) - 8 \times x = 0$

$\Rightarrow M = 2x + 12$





Free body diagram

Region DB ($4 < x < 6$):

$\uparrow \sum F = 0$

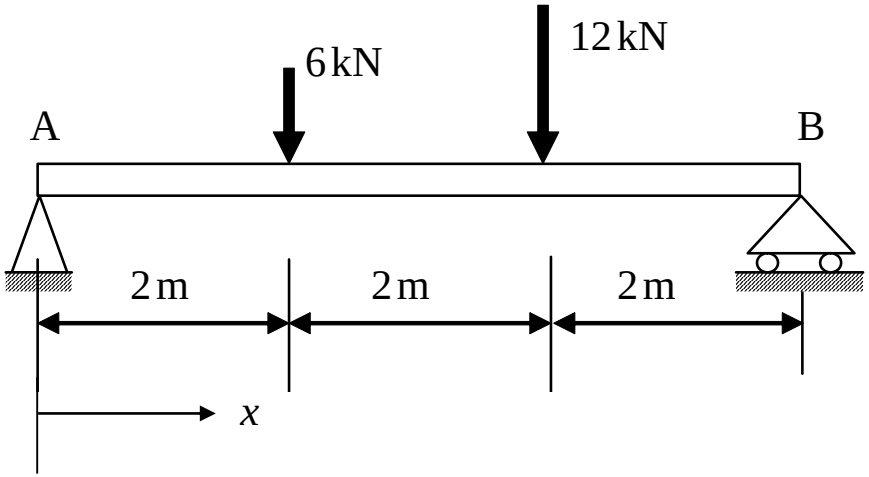
$\Rightarrow 8 - 6 - 12 - V = 0$

$\Rightarrow V = -10\,\text{kN}$

$\circlearrowleft \sum M_{\text{ about location } x} = 0$

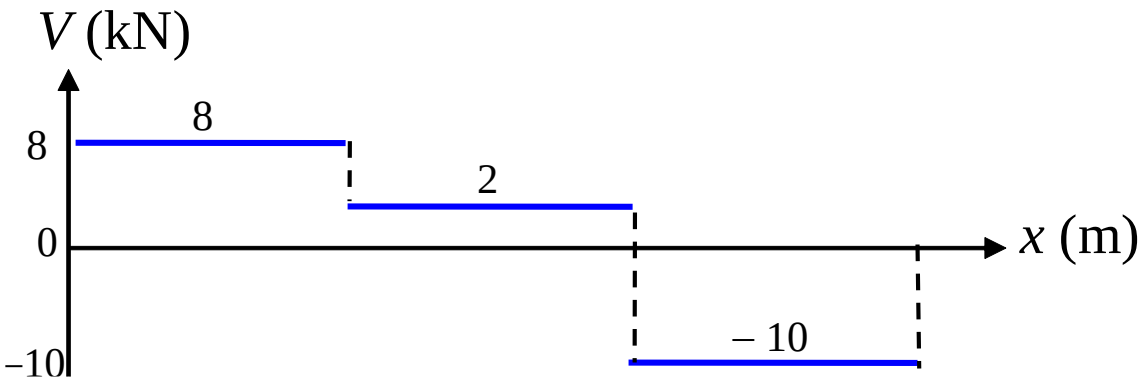
$\Rightarrow M + 12\times(x-4) + 6\times(x-2) - 8\times x = 0$

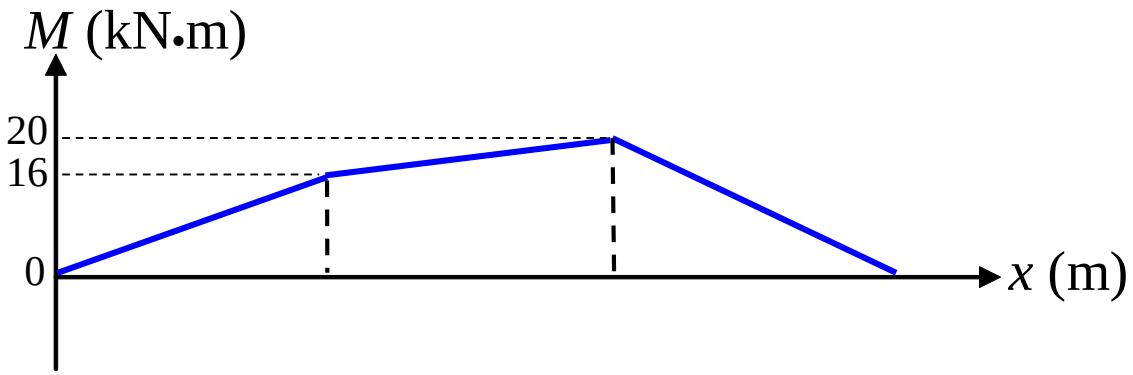
$\Rightarrow M = -10x + 60$



3. Draw the diagrams from left to right regions.

Abrupt change in shear force occurs at any point where a concentrated load acts. Shear force decreases suddenly by *P* (downward) as we move from left to right through a point of application of a force *P*.





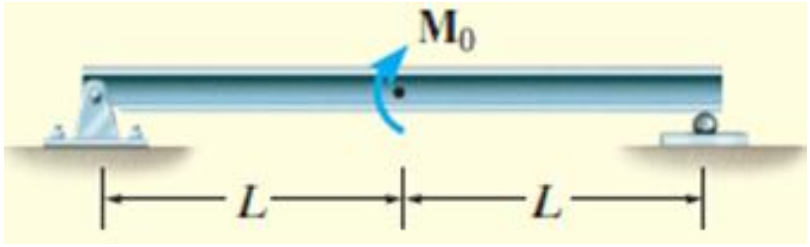
Solution procedure for constructing BM and SF diagrams

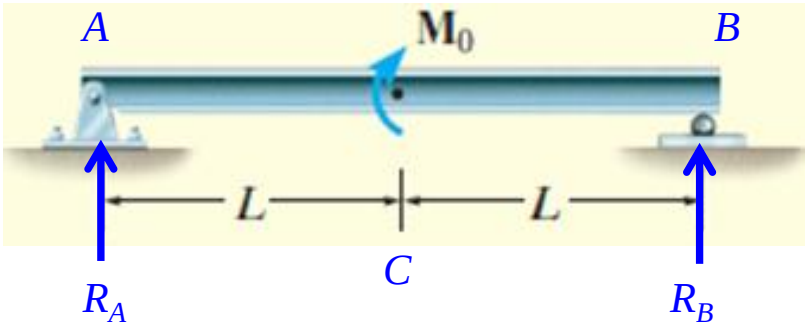
1. Determine the support reaction from the equilibrium of the entire beam.

2. The beams are divided into several regions. Using method of free-body diagram and calculate the internal shear forces and moments for each region (expressed as a function of x).

3. Draw the diagrams from left to right regions.

Example 2: Draw the shear force and bending moment diagrams for the beam shown.





Note: We draw the unknown external force R_A in the positive direction (upward). The negative value we obtain indicate that it acts downward.

Solution:

1 Find the support reactions

$$\circlearrowleft \sum M_{\text{aboutpoint } A} = 0$$

$$\Rightarrow R_B \times (2L) - M_0 = 0$$

$$\Rightarrow R_B = \frac{M_0}{2L}$$

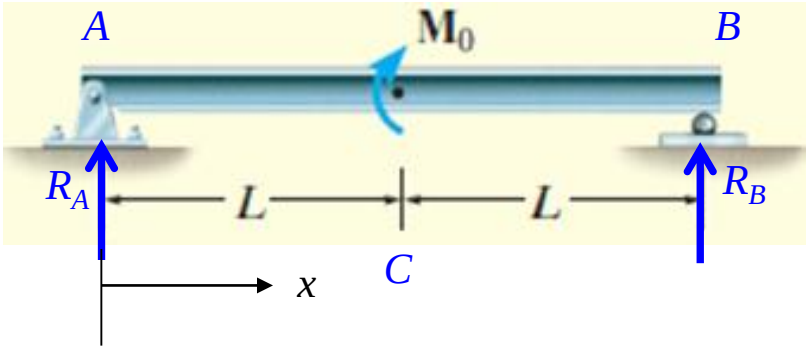
$$\circlearrowleft \sum M_{\text{aboutpoint } B} = 0$$

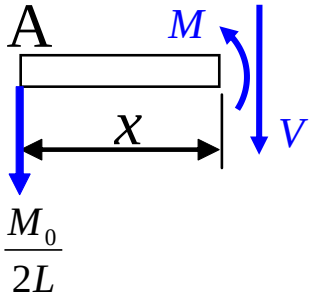
$$\Rightarrow -R_A \times (2L) - M_0 = 0$$

$$\Rightarrow R_A = -\frac{M_0}{2L}$$

2 Divide the beam into 2 regions based on the loading conditions: AC & CB.

Express the SF and BM for each region





Free body diagram

Note: when you study a free body, V and M act as external forces for this free body.

Region AC ($0 < x < L$):

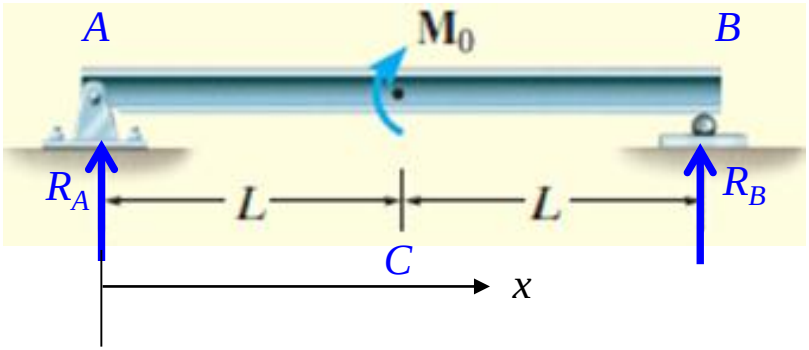
$\uparrow \sum F = 0$

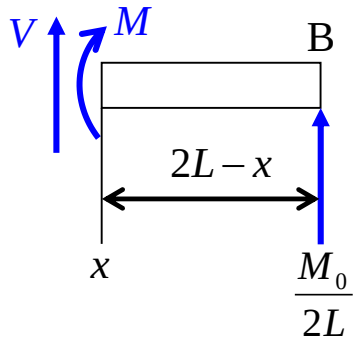
$\Rightarrow \quad -V - \frac{M_0}{2L} = 0 \quad \Rightarrow \quad V = -\frac{M_0}{2L}$

$\circlearrowleft \sum M_{\text{about location } x} = 0$

$\Rightarrow \quad M + \frac{M_0}{2L} x = 0$

$\Rightarrow \quad M = -\frac{M_0}{2L} x$





Free body diagram

Region CB ($L < x < 2L$):

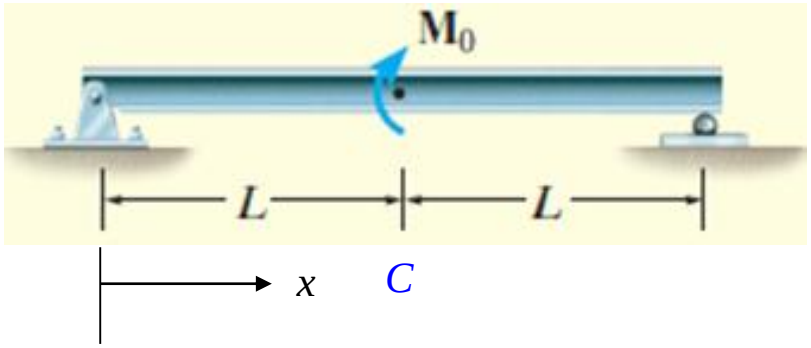
$\uparrow \sum F = 0$

$\Rightarrow V + \frac{M_0}{2L} = 0 \quad \Rightarrow V = -\frac{M_0}{2L}$

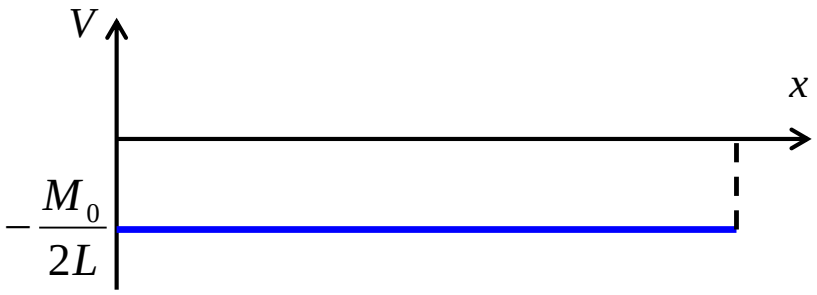
$\circlearrowleft \sum M_{\text{about location } x} = 0$

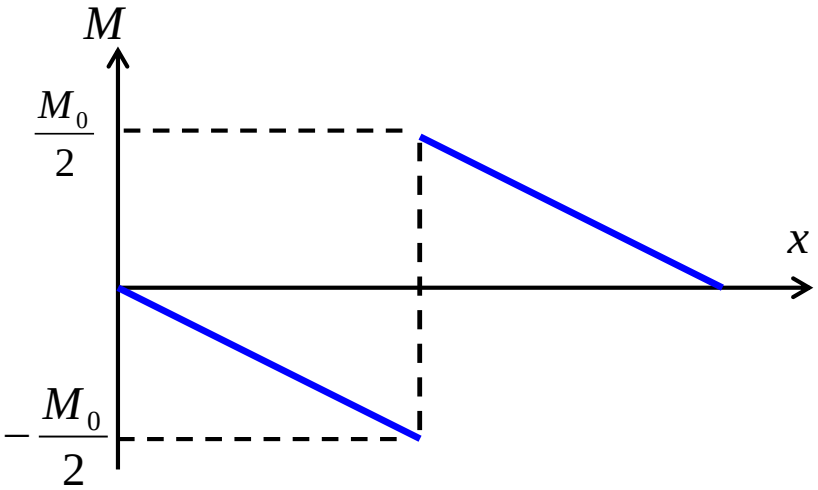
$\Rightarrow -M + \frac{M_0}{2L}(2L - x) = 0$

$\Rightarrow M = \frac{M_0}{2L}(2L - x)$



3. Draw the diagrams from left to right regions.



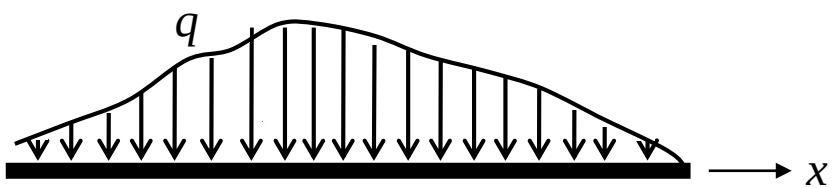


Abrupt change in bending moment occurs at any point where a couple acts. Bending moment increases suddenly by M_0 (clockwise) as we move from left to right through a point of application of a couple M_0.

6.4 Distribution forces

It is rare to find a force applied at a single point. Indeed if a force were truly applied at just one point, the area over which it was applied would be infinitesimally small and the stress in the region of its application would approach infinity.

It is much more common to find forces distributed over a finite area.



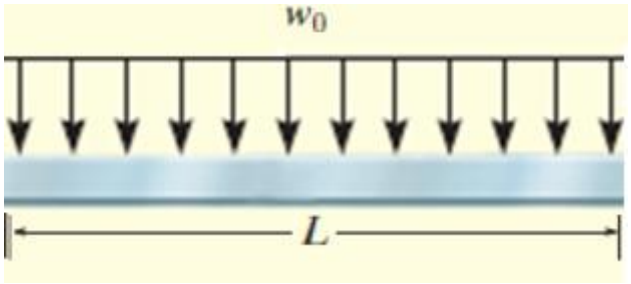
In this chapter we look at forces distributed along beams. The intensity of the distributed force can be constant or can vary along the beam, and is given the symbol q.

q is defined to be positive if it is acting downward, and it has units of force per length (N/m). q can be a function of distance along the beam (x).

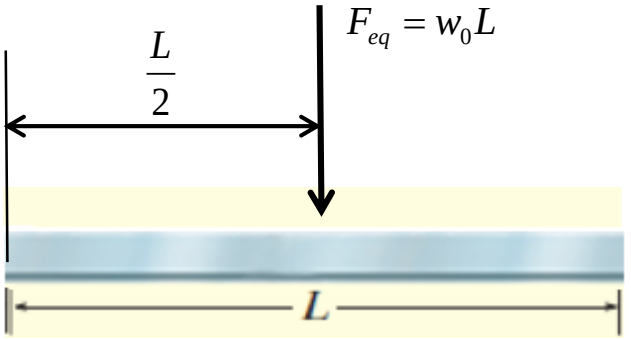
As we saw in the unit *Fundamentals of Mechanics*, a distribution force can be replaced by an equivalent point force (F_{eq}), which will have exactly the same **external** effects (effects at any point outside the region where the force is applied).

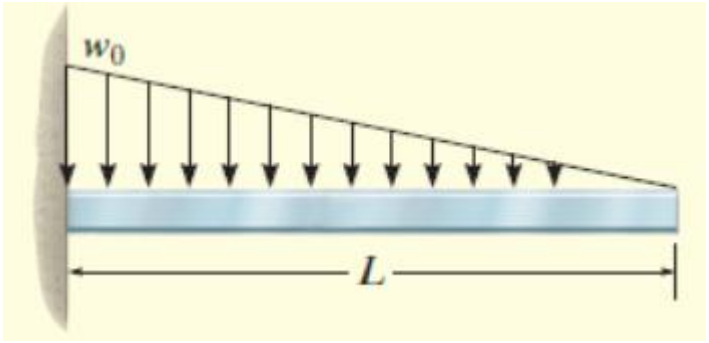
The equivalent point force F_{eq} has a magnitude equal to the area of the region under a graph of q versus x. The line of action of F_{eq} passes through the centroid of this region.

Two common examples of distributed forces are:

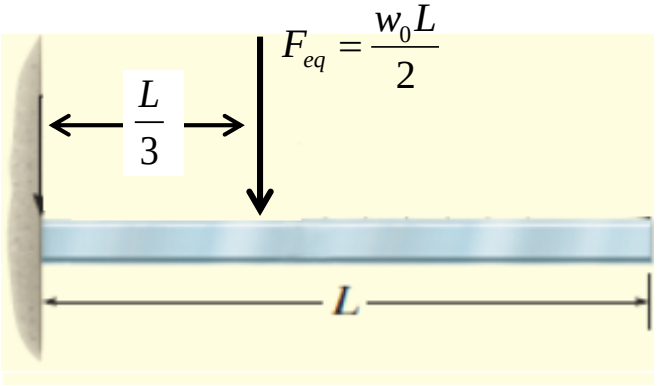


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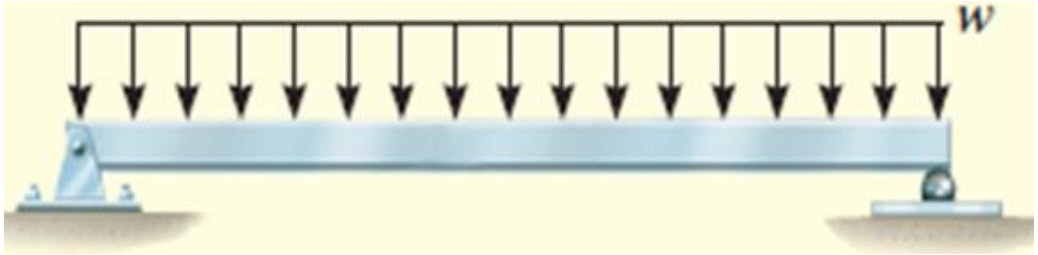




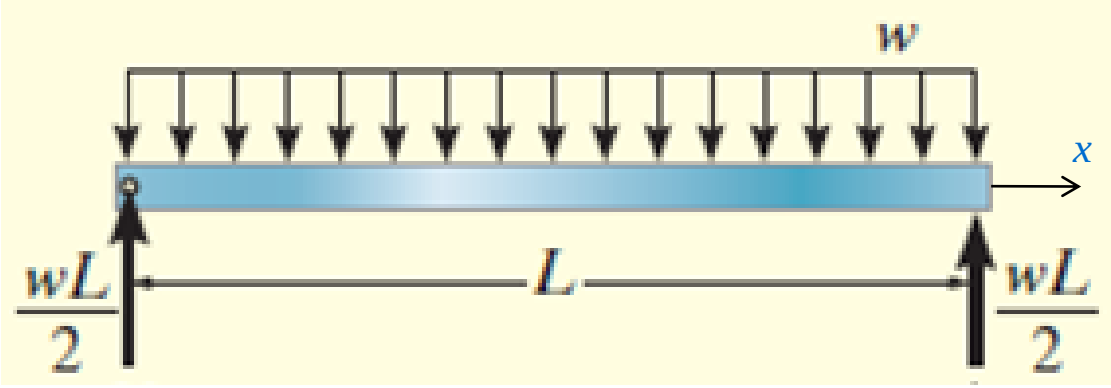
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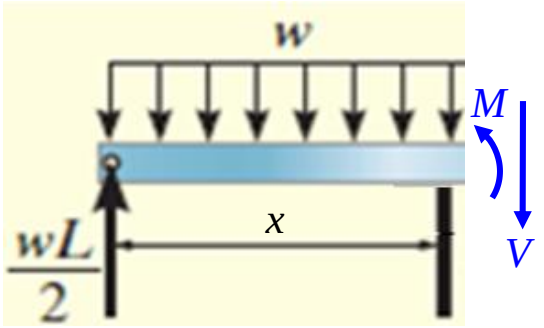


Example 3: Draw the shear and moment diagrams for the beam shown below. Find the maximum value of the bending moment.

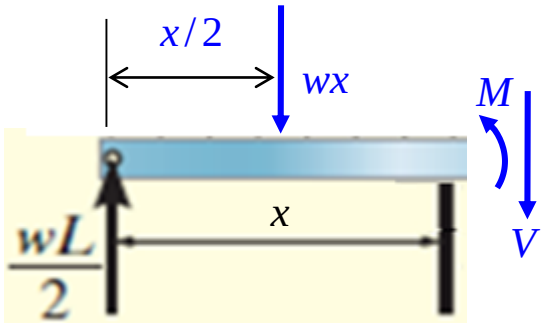


Solution: Due to symmetry, the support reactions R_a and R_b at both ends of the beam are equal and are $R_a = R_b = wL/2$. Since the beam is simply supported at both ends, the moments are zero at both ends.

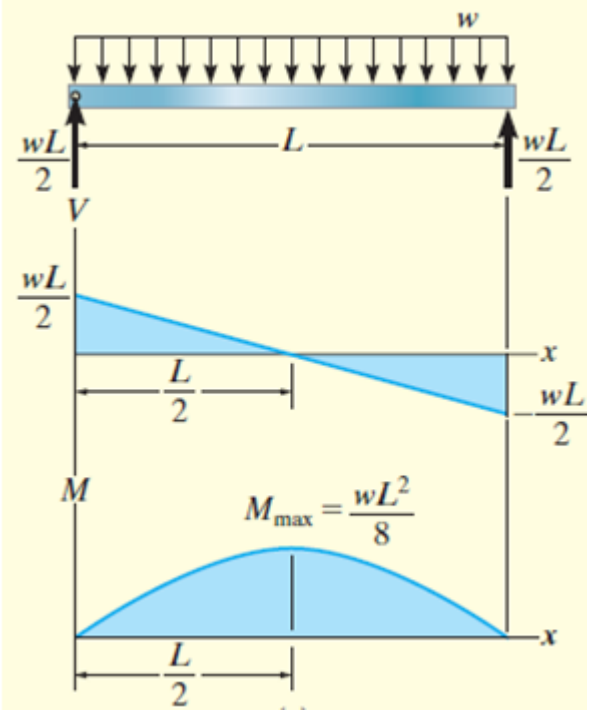




Free body diagram



Equivalent problem



$$\uparrow \sum F = 0$$

$$\Rightarrow \quad -V - wx + \frac{wL}{2} = 0$$

$$\Rightarrow \quad V = \frac{wL}{2} - wx$$

$$\circlearrowleft \sum M_{\text{aboutlocation}x} = 0$$

$$\Rightarrow \quad M + wx \times \frac{x}{2} - \frac{wL}{2} \times x = 0$$

$$\Rightarrow \quad M = \frac{w}{2}\left(Lx - x^2\right)$$

- The point of maximum bending moment can be found through

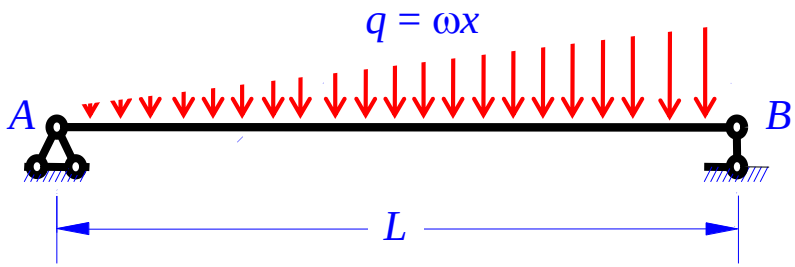
  $dM/dx = 0$ or

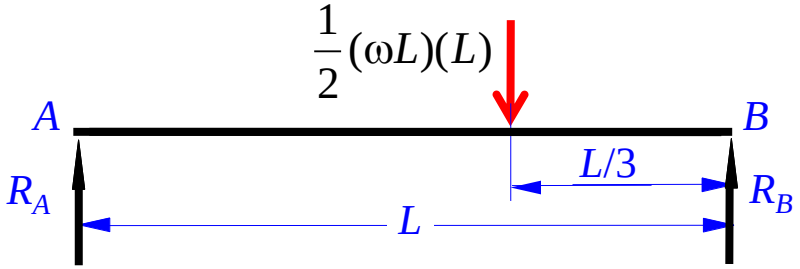
$$V = 0 \Rightarrow x = \frac{L}{2}$$

- This value of x represents the point on the beam where the *maximum moment* occurs

$$M_{max} = \frac{w}{2}\left[L\frac{L}{2} - \left(\frac{L}{2}\right)^2\right] = \frac{wL^2}{8}$$

Example 4: Obtain the shear force and moment for the beam shown. Suppose the applied distribution load is ωx.





Free body diagram

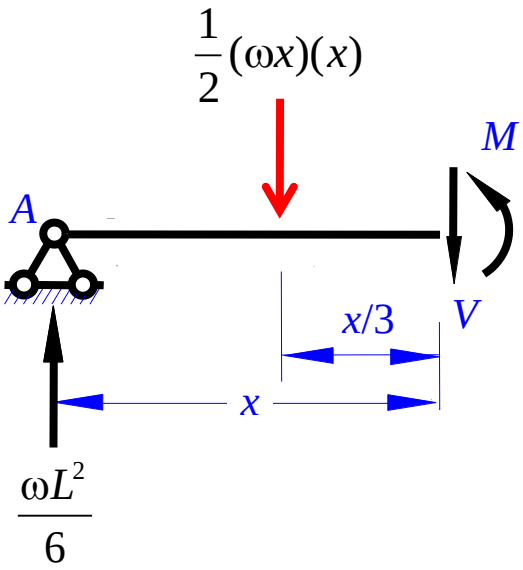
Solution:

1 Find the support reactions

$$\circlearrowleft \sum M_B = 0$$

$$\Rightarrow -R_A L + \frac{\omega L^2}{2} \times \frac{L}{3} = 0 \Rightarrow R_A = \frac{\omega L^2}{6}$$

2 Find the shear force and bending moment by using FBD



Free body diagram

$\uparrow \sum F = 0$

$\Rightarrow \frac{\omega L^2}{6} - V - \frac{\omega x^2}{2} = 0$

$\Rightarrow V = -\frac{\omega x^2}{2} + \frac{\omega L^2}{6}$

$\circlearrowleft \sum M_{\text{about location } x} = 0$

$\Rightarrow -\frac{\omega L^2}{6} \times x + M + \frac{\omega x^2}{2} \times \frac{x}{3} = 0$

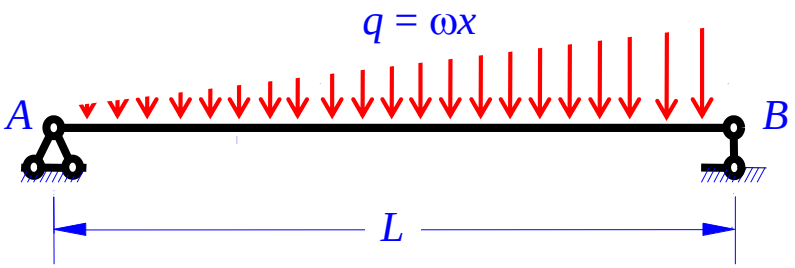
$\Rightarrow M = \frac{\omega L^2 x}{6} - \frac{\omega x^3}{6}$

6.5. Relationship between external forces (couples) and BM & SF

In example 4 we found the following expressions for shear force and bending moment in the beam shown.

$$V = -\frac{\omega x^2}{2} + \frac{\omega L^2}{6}$$

$$M = \frac{\omega L^2 x}{6} - \frac{\omega x^3}{6}$$



Is there a relationship between q, V and M ?

By differentiating V and M, with respect to x, we see that

$$\frac{dV}{dx} = -q \qquad (6.1)$$

$$\frac{dM}{dx} = V \qquad (6.2)$$

$$\frac{d^2M}{dx^2} = -q \qquad (6.3)$$

These are very useful formula for determining expressions V and M .

6.7 Note:

1. Make a cut and a new free body diagram for every change in the beam's loading.

2. When determining the shear force and bending moment under a distributed force, we can cut the beam and replace the distributed force by an equivalent point force.

3. We could use a free body diagram of section of the beam on the right-hand side of the cut point instead of the left-hand side, as used in examples 2 and 5.

   a) Since every action has an equal and opposite reaction, the positive directions for V and M on the right-hand portion are equal and opposite to those on the left-hand portion.
   b) The length of the portion is equal to the overall length minus x.

4. In text books and literature, a lot of other methods for determining the shear forces and bending moments in the beam have been introduced. Based on experience, the simple and easy to understand method of FBD is most important.

Quick review

Bending moment and shear force

BM & SF Diagrams by using free body diagram



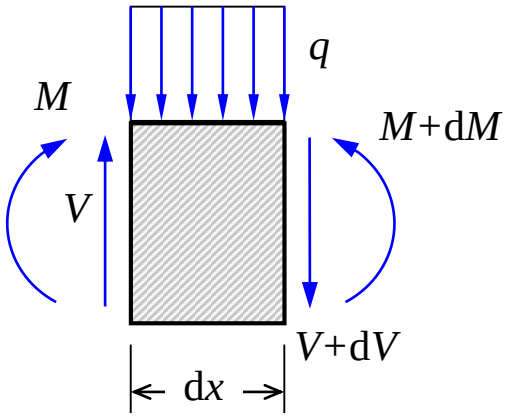
$\uparrow \sum F = 0$, $\circlearrowleft \sum M = 0$

BM & SF Diagrams by solving differential equations

$$\frac{dV}{dx} = -q$$

$$\frac{dM}{dx} = V$$

$$\frac{d^2M}{dx^2} = -q$$



Note: Arrows show the positive sign convention for the forces.